Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device).*
The RS485 Digital IO module is sturdy, has low power usage and is easy to use.

4 Port DI Module: -

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw Terminal Block
Dimensions	60 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 – 24 VDC or 24 V AC/DC Typical – 12V DC @ 80mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	RoHS, APPROVED CE APPROVED

DI Inputs –

Channels	4
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic ‘0’ = <1 VDC, Logic ‘1’ = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Additional Features: -

All inputs and communication port isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

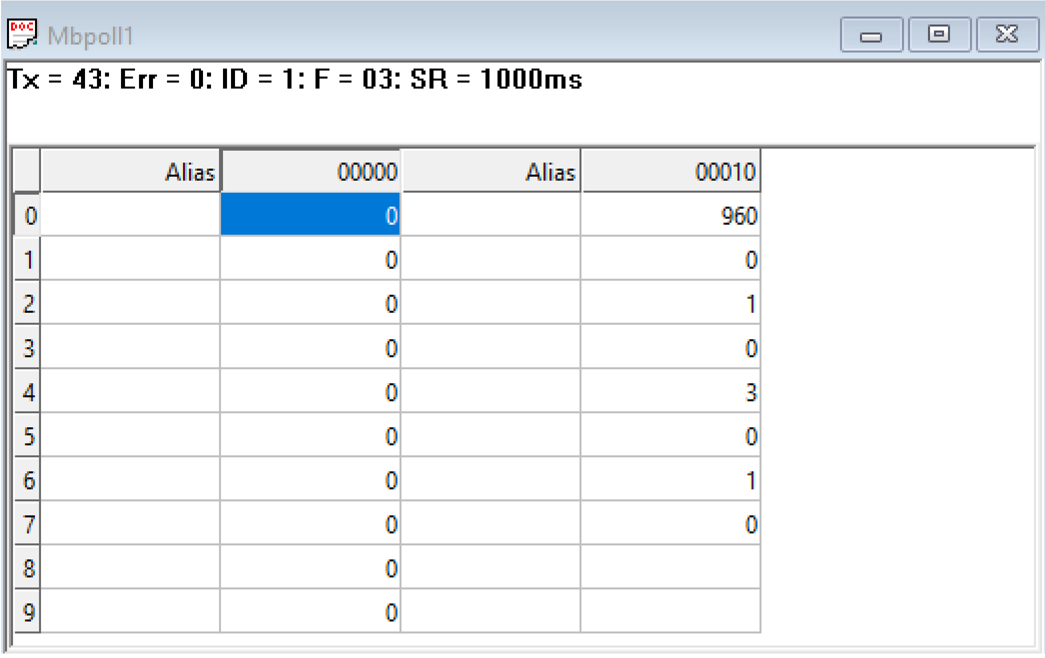
Configuration Settings: -

Communication Speed	9600 – 19200 Kbps (DIP SW selectable)
Data Bits	8
Parity	None
CRC	Yes
Stop bit	1
Slave Id	Soft Setting.
Function code DI	0x02 Read discrete Input
DI Register Address	0,1,2,3.
Function code For Soft Serial	0x03 Read Holding Register
Soft Serial Register Address	10,11,12,13,14,15,16,17.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	DI 1(With & Without Potential)	10001	0X02	RS485	1 Bit Boolean
1	DI 2(With & Without Potential)	10002	0X02	RS485	1 Bit Boolean
2	DI 3(With & Without Potential)	10003	0X02	RS485	1 Bit Boolean
3	DI 4(With & Without Potential)	10004	0X02	RS485	1 Bit Boolean
4	Display Baud Rate (Default:960)	40011	0X03	RS485	16 Bit Unsigned int
5	Enter New Baud Rate	40012	0X03	RS485	16 Bit Unsigned int
6	Display Slave ID (Default:1)	40013	0X03	RS485	16 Bit Unsigned int
7	Enter New Slave ID	40014	0X03	RS485	16 Bit Unsigned int
8	Display Parity (Default: None-3, Odd-1, Even-2)	40015	0X03	RS485	16 Bit Unsigned int
9	Enter New Parity	40016	0X03	RS485	16 Bit Unsigned int
10	Display Stop Bit (start-2/stop bit-1)	40017	0X03	RS485	16 Bit Unsigned int
11	Enter New Start/ Stop Bit	40018	0X03	RS485	16 Bit Unsigned int

1. In register 40015- you will able to see the current parity and you can change it in register 40016
2. In register 40017- you will able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc. Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.

BAUD RATE and SLAVE ID DESCRIPTION:

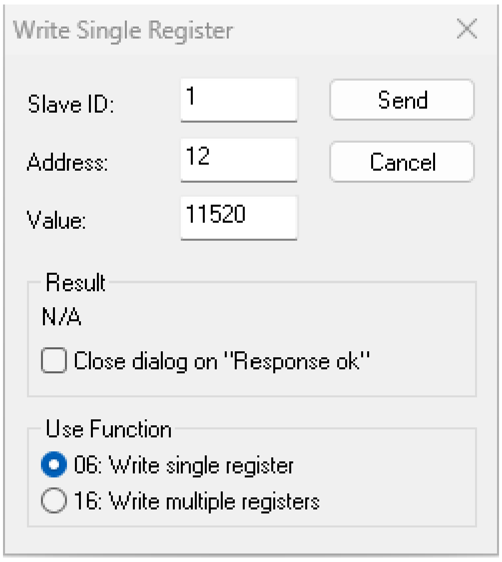

Mbpoll1

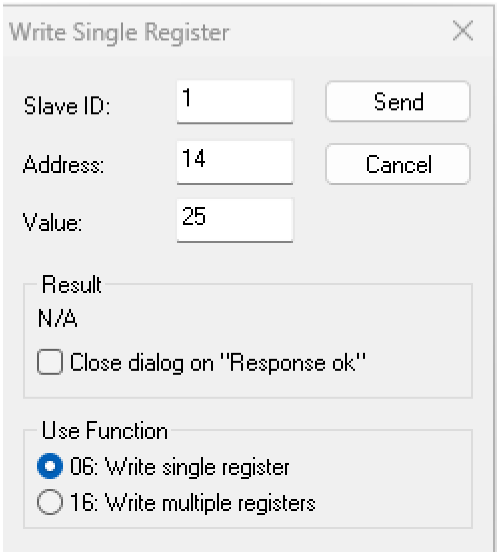
Tx = 43: Err = 0: ID = 1: F = 03: SR = 1000ms

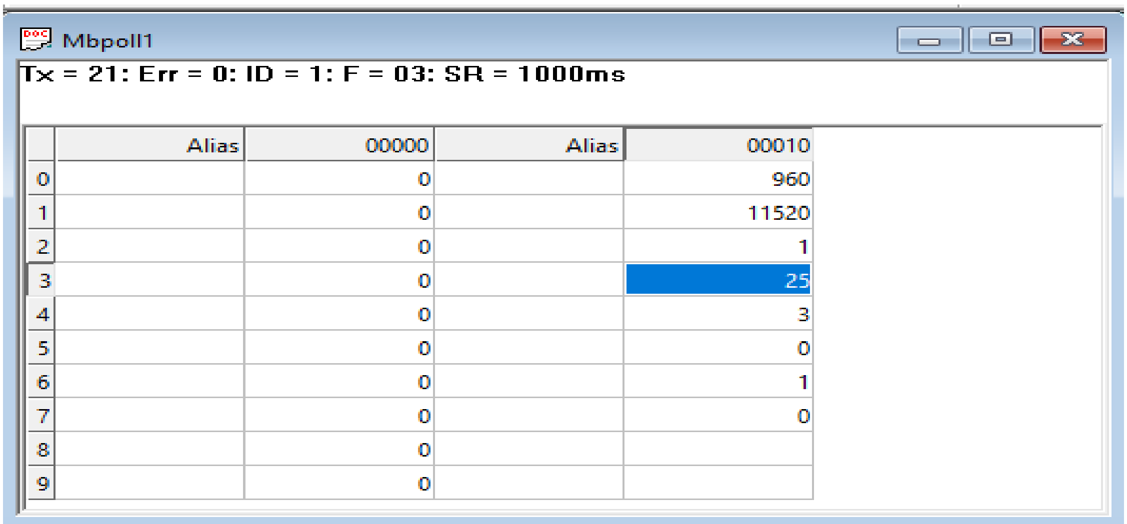
	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.

To change the baud rate and slave id you have to enter a value on the 40012 resistor for the baud rate and 40014 for the slave id as shown in the image below.






Mbpoll1

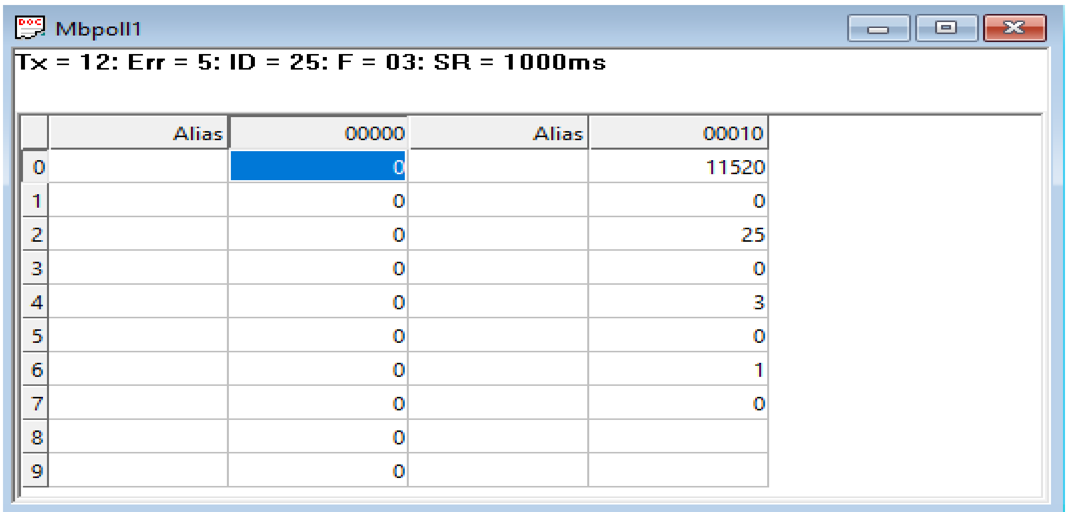
Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:

1. **If you want to enter the baud rate 9600/19200/38400/57600/115200etc.**
   Enter it like this 960/1920/5760/11520etc.
2. **After that press the reset button or reboot the system.**
3. **Then you will get the present value of 11520 at 40011 baud rate(115200) and 25 at 40013 slave id.**
4. **You can Change Parity Bit , stop bit Of the device with same way.**


Mbpoll1

Tx = 12: Err = 5: ID = 25: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		11520
1		0		0
2		0		25
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

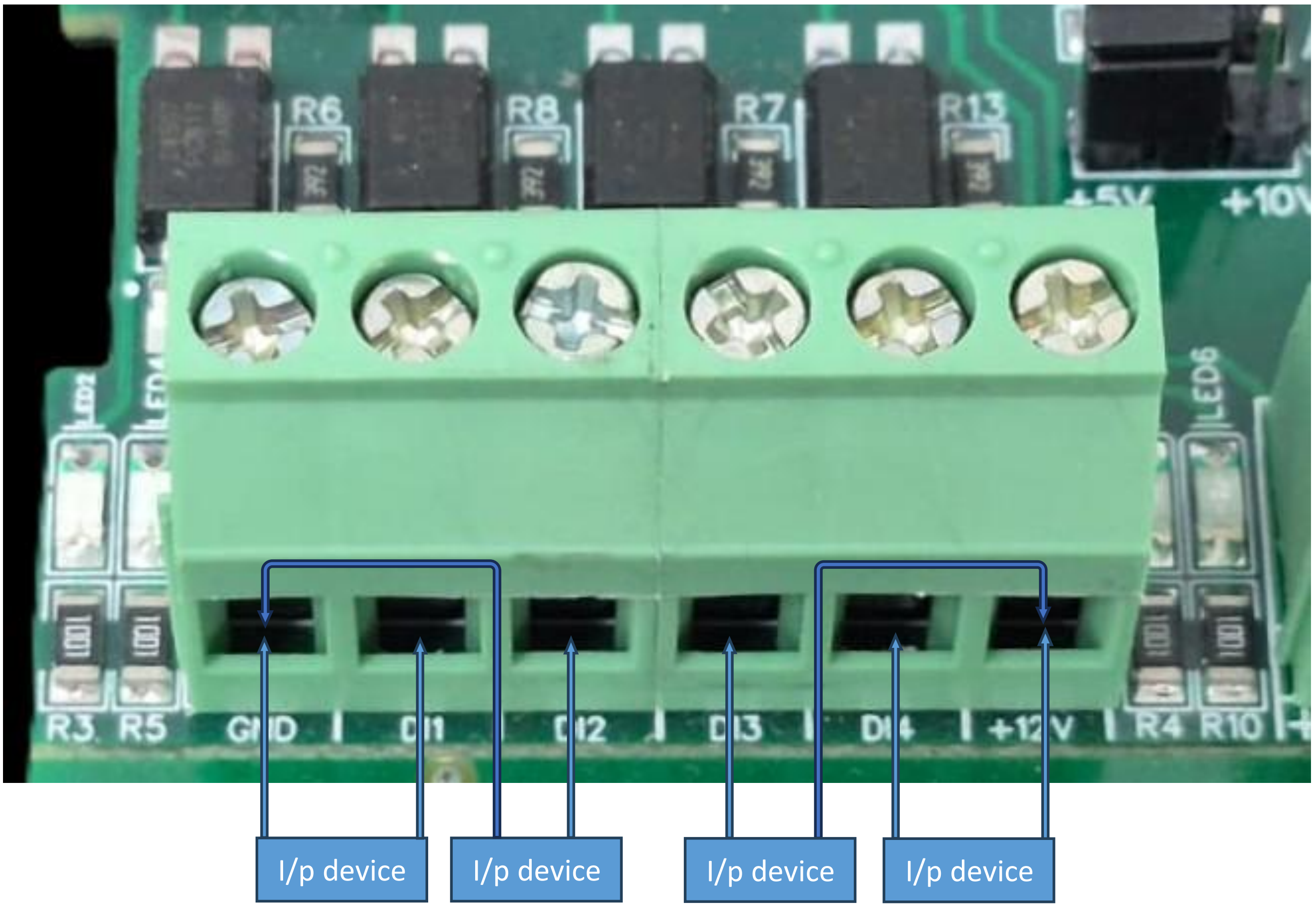
22 AWG Cable	Shielded and twisted pair
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100

Soft Setting Description

Default Setting for Dip Switch

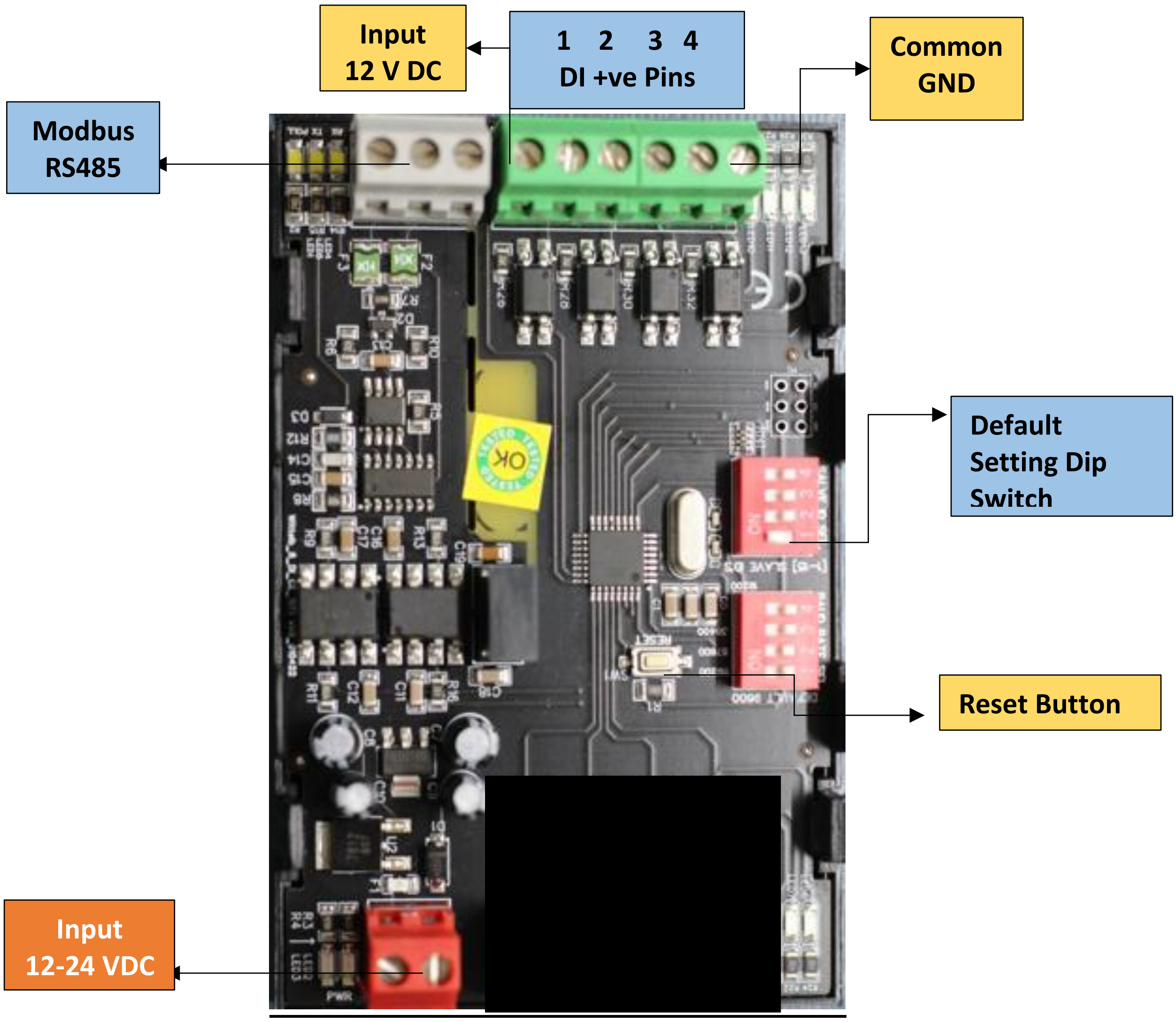
NOTE – By default the Jumper setting on the board is 10 V and 10 Bit.
– By default, Baud rate DIP Switch 1 is set to ON, toggling it OFF and Pressing reset button will bring the device to default RS485 Configuration i.e. Baud rate - 9600, Stop bit - 1, Parity - None, Data Bit - 8. Keep the indicated DIP switch to ON position for Setting desired configuration.

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
  - **GND:** Common Ground reference.
  - **DI1 – DI4:** Individual Digital Input channels.
  - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
  - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
  - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

For Digital Input with Potential (3V to 30V) Contact + Common GND terminal should be connected on the Board, e.g DI1 +ve terminal and GND pin.



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